

Reviewer Pack — Minimal Rank of p-Adic Differentials over AC^0 Parity via Hod...

Entry 86cbcf58487a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of p-Adic Differentials over AC^0 Parity via Hodge Decomposition

Entry ID: 86cbcf58487a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 20:46:35 UTC

1. Conjecture

Field A (mathematical branch): p-adic Analysis (Differential Equations) **Field B** (complexity object): Complexity Theory: AC^0 Parity

Statement:

{‘quantitative_relation’: ‘ $E[\rho(H(n))] = \Theta(\log^2(n))$ for all AC^0 parity circuits C with n inputs, where $\rho(H(n))$ is the Hodge rank of the p-adic differential form H associated with C .’, ‘falsifier_formulation’: ‘There exists an AC^0 parity circuit C with n inputs such that the Hodge rank of its associated p-adic differential form is less than $\Theta(\log^2(n))$.’}

Rationale (proposer’s reasoning):

{‘explanation’: ‘p-adic analysis provides a framework for studying arithmetic properties of functions, which can be applied to analyze the complexity of Boolean functions. The Hodge decomposition in p-adic geometry may reveal hidden structures in AC^0 circuits that are not apparent through traditional methods.’, ‘potential_structure’: ‘This conjecture could expose new connections between arithmetic properties of functions and their computational complexity, potentially leading to improved lower bounds for AC^0 parity.’}

Taxonomy category: AC^0_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b646638caf4e56e3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the empirical average Hodge rank over 30 random seeds is within $\pm 50\%$ of $\Theta(\log^2(n))$, otherwise it is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=-9, elapsed=93.4s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_ac0_parity_circuit(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def p_adic_differential_form(circuit):
        n = len(circuit)
        H = [[0] * n for _ in range(n)]
        for i in range(n):
            for j in range(i+1, n):
                if circuit[i] != circuit[j]:
                    H[i][j] = 1
                    H[j][i] = 1
        return H

    def hodge_rank(H):
        n = len(H)
        rank = 0
        for i in range(n):
            if all(H[j][i] == 0 for j in range(n) if j != i):
                rank += 1
        return rank

    def log2(x):
        return math.log2(x) if x > 0 else float('-inf')

    n_values = [5, 10, 15, 20, 30, 40]
    hodge_ranks = []

    for n in n_values:
        circuit = generate_ac0_parity_circuit(n)
        H = p_adic_differential_form(circuit)
```

```

    rank = hodge_rank(H)
    hodge_ranks.append(rank)

avg_hodge_rank = sum(hodge_ranks) / len(hodge_ranks)
expected_avg = sum(log2(n)**2 for n in n_values) / len(n_values)

return {
    "metric_name": "Hodge Rank",
    "metric_value": avg_hodge_rank,
    "instances_tested": len(n_values),
    "conjecture_holds": abs(avg_hodge_rank - expected_avg) <= 0.5 * expected_avg,
    "counterexample": "" if avg_hodge_rank >= expected_avg else f"Expected Hodge rank {expected_avg}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    avg_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={avg_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={avg_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        counterexample = next(r["counterexample"] for r in results if r["counterexample"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

(empty)

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means that the empirical average Hodge rank could not be calculated to verify the conjecture. | next: Re-run the test with a different set of seeds or parameters to ensure it completes successfully and provides the necessary data for verification.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	?	?	0	0	15140
2	propose	ollama_remote	glm4:latest	0	0	20640
3	preregistration	ollama_remote	glm4:latest	0	0	5817
4	novelty	ollama_remote	glm4:latest	0	0	5073
5	novelty	ollama_remote	glm4:latest	0	0	5415
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12841
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8997
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8388
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9710
10	judge	ollama_remote	glm4:latest	0	0	57171

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 149193 ms total latency. Provider mix: {'?': 1, 'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/86cbcf58487a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/86cbcf58487a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/86cbcf58487a.tar.gz` (if generated)